



**TÉCNICO**  
LISBOA

## **Exercise I**

# **Network Planning - Design**

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## Strategic Context



- EU and UN have launched coordinated efforts to reduce greenhouse gas (GHG) emissions, setting clear targets to **achieve carbon neutrality by 2050**
- **Green hydrogen** has been identified as a **key enabler for decarbonizing various sector**, including the energy and industry sectors

## Green Hydrogen Role



- Emerging as **energy carrier, alternative fuel, and industrial feedstock**
- Produced from **renewable energy sources (RES)**
- Supports the **decarbonization** of the industrial sector, particularly **hard-to-abate industries** (such as heavy manufacturing) , through **integration with renewable energy systems**

## Integration Challenges and Supply Chain Focus



- The hydrogen supply chain currently **lacks a well-defined value chain and robust design and planning**
- Key challenges include **reducing overall costs** and **optimizing infrastructure**, particularly considering **industrial-scale readiness and maturity**

# The Green Hydrogen Supply Chain (GHSC)

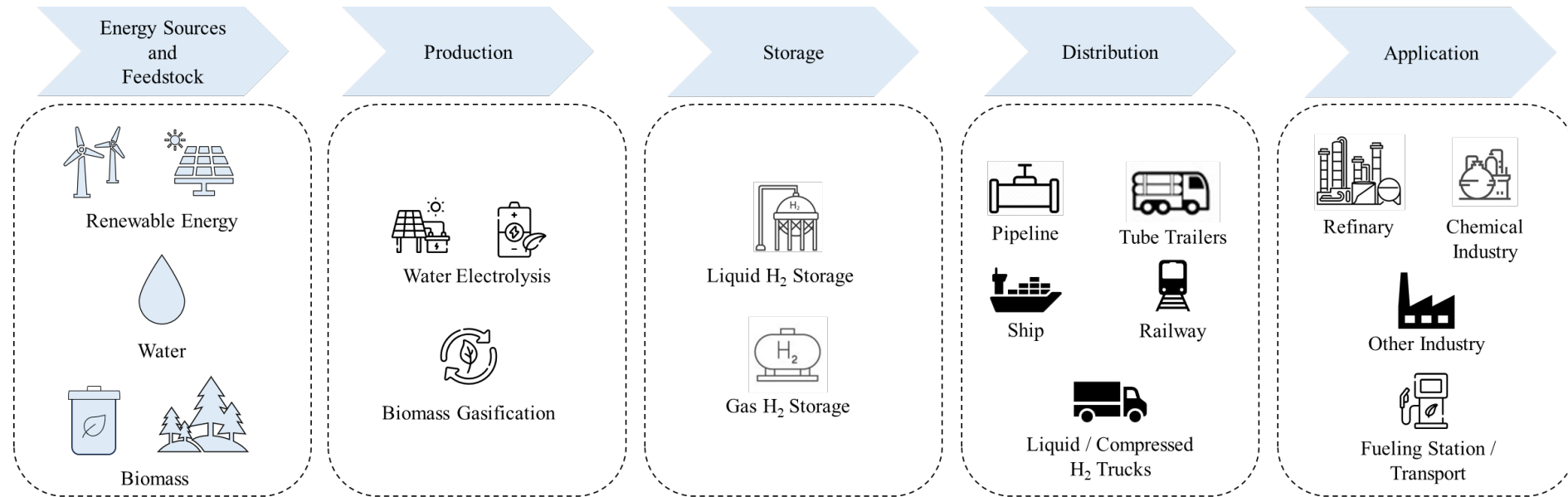


Figure I. Overview of the GHSC superstructure

- 1) Formulate a mathematical model, that represents the green hydrogen supply chain network, and which can be used to optimize the network design while minimizing total costs;
- 2) The green hydrogen supply chain is applied in Portugal, with a particular focus on meeting H<sub>2</sub> demand for the country's industrial sector;
- 3) The goal is to develop a robust and efficient supply chain capable of addressing this demand, while tackling key challenges such as high infrastructure costs and evolving market penetration.

- 4) The model should optimize the GHSC network design and planning, while minimizing the total supply chain costs. This should be accomplished by determining:
- the **number, location, and size of production plants and storage facilities;**
  - **allocation of resources and product flows;**
  - **number of trailers** required for transportation from production plants to storage facilities, and from storage facilities to consumers.

## Mathematical models are composed by:

**Variables:** decisions to be taken on the system and quantities to calculate.

**Operators:** act on these variables, which can be algebraic operators (**parameters**), functions, differential operators, etc.

**Indices:** identify entities in the system.

If the objective functions and constraints are represented entirely by linear equations, then the model is regarded as a **linear model**.

If one or more of the objective functions or constraints are represented with a nonlinear equation, then the model is known as a **nonlinear model**.

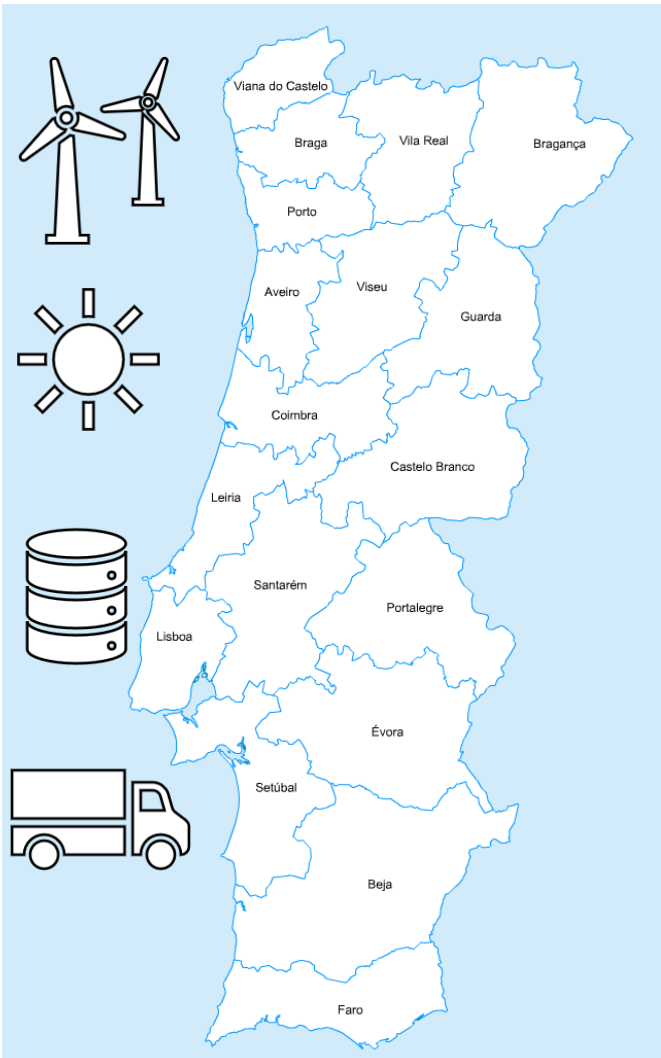
1. What is the system in study?
2. What are the different entities involved in the system? How many indices do we need?
3. What is the available data? How many parameters do we need?
4. What is the unknown data? How many variables do we need?
5. What are the costs involved in this supply chain? How many terms do we need for the objective function?
6. Do we have constraints? How many?

- The mathematical formulation of the problem under study should establish the parameters and the decision variables, which correspond to the variables expected to be estimated as an output of the model solution.



- The developed model should find values for the decision variables, that fulfil all constraints and respect the parameters, while optimizing the objective function (minimize the GSCH network costs).

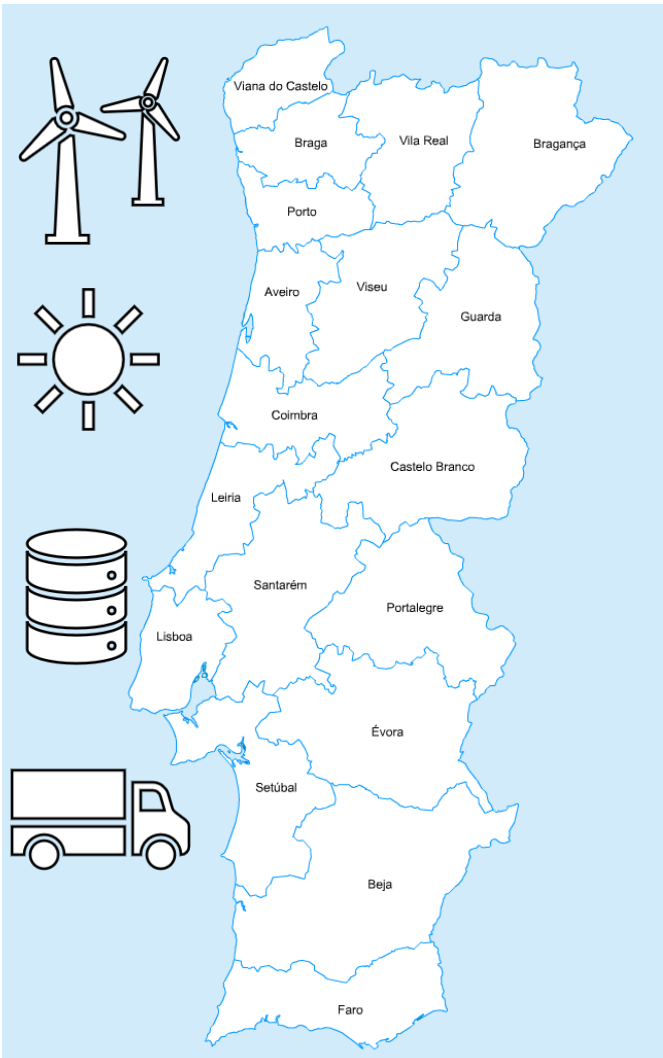




## To develop the model, the following data is required (inputs):

- Location and capacity of energy source suppliers
- Capacity of H<sub>2</sub> production and storage facilities
- Capital and operating costs for each production and storage technology
- Shipping costs of the considered transportation mode
- Distances between locations
- Locations of demand points and respective product demand
- Time horizon considered for the study

# Mathematical Model Formulation – indices



## Indices

$e$  – energy source types

(wind and solar)

$i, j$  – locations

(18 districts)

$l$  – production plant type

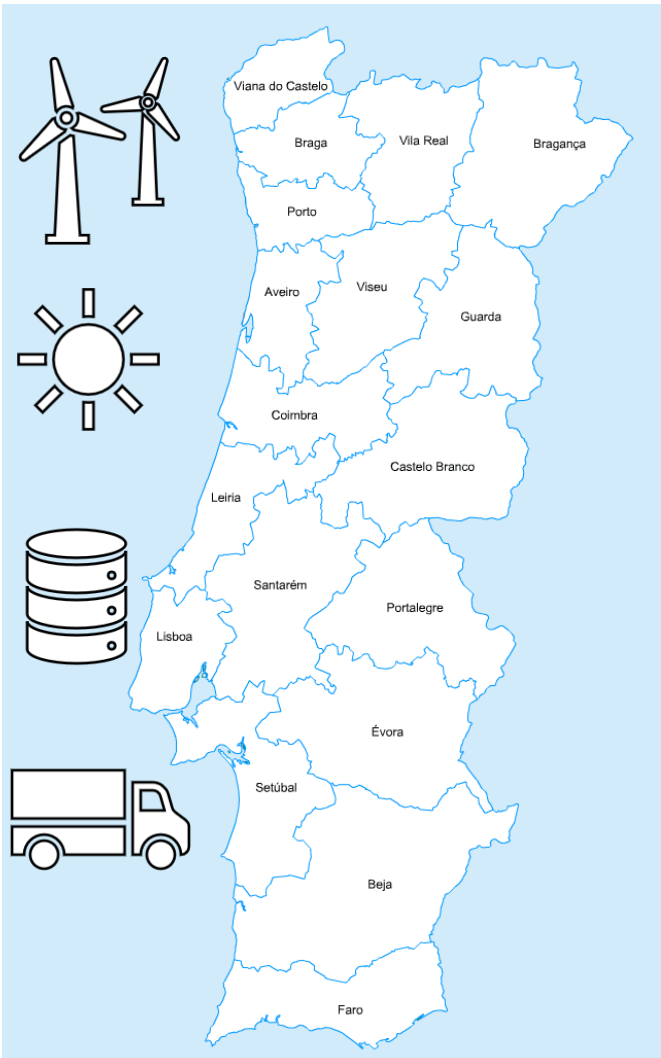
(small, medium, large)

$m$  – storage facility type

(tank 1, tank 2, tank 3, tank 4, tank 5)

$t$  – periods

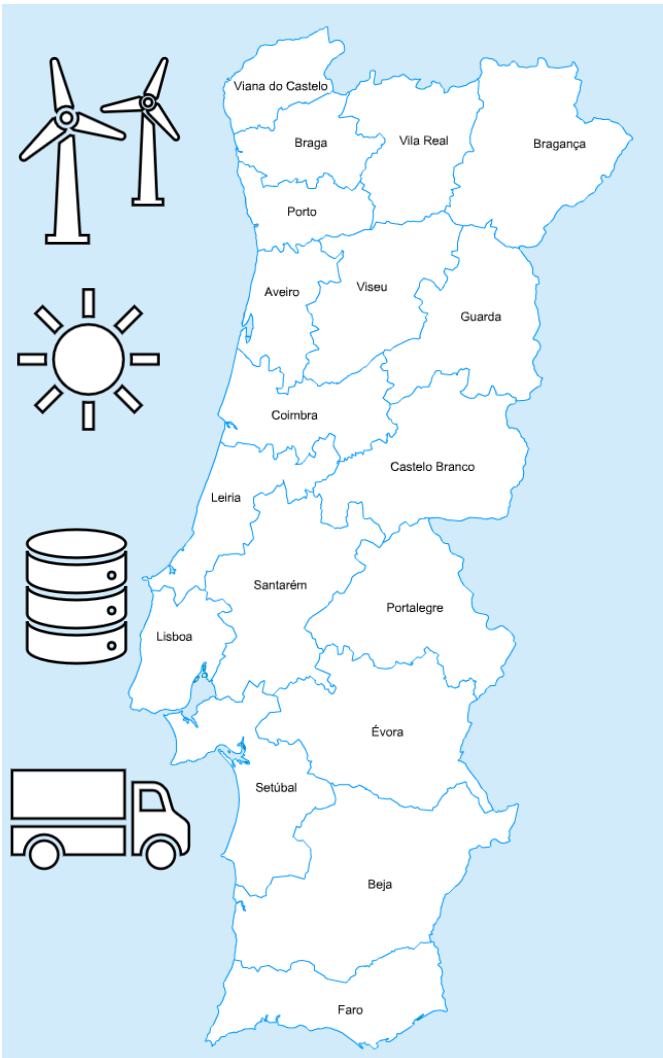
# Mathematical Model Formulation – Parameters



## Parameters

|               |                                                                                                                              |
|---------------|------------------------------------------------------------------------------------------------------------------------------|
| $\alpha_t$    | Penetration rate of H <sub>2</sub> at period $t$                                                                             |
| $\gamma_{el}$ | Energy conversion factor (amount of energy source $e$ to produce 1 ton of H <sub>2</sub> , in production plant of type $l$ ) |
| $b_i$         | H <sub>2</sub> demand at location $l$                                                                                        |
| $c^T$         | Shipping cost of H <sub>2</sub>                                                                                              |
| $dist_{ij}^T$ | Driving distance from location $i$ to locations $j$                                                                          |
| $I_l^P$       | Capital cost of a production plant of type $l$                                                                               |
| $I_m^S$       | Capital cost of a storage facility of type $m$                                                                               |
| $ESCost$      | Renewable energy sources cost                                                                                                |
| $ESAV_{ie}$   | Amount of available energy source $e$ at location $i$                                                                        |
| $o_l^P$       | Operating cost of a production plant of type $l$                                                                             |
| $o_m^S$       | Operating cost of a storage facility of type $m$                                                                             |
| $u_l^P$       | Production capacity of a production plant of type $l$                                                                        |
| $u_m^S$       | Capacity of a storage facility of type $m$                                                                                   |

# Mathematical Model Formulation – Variables



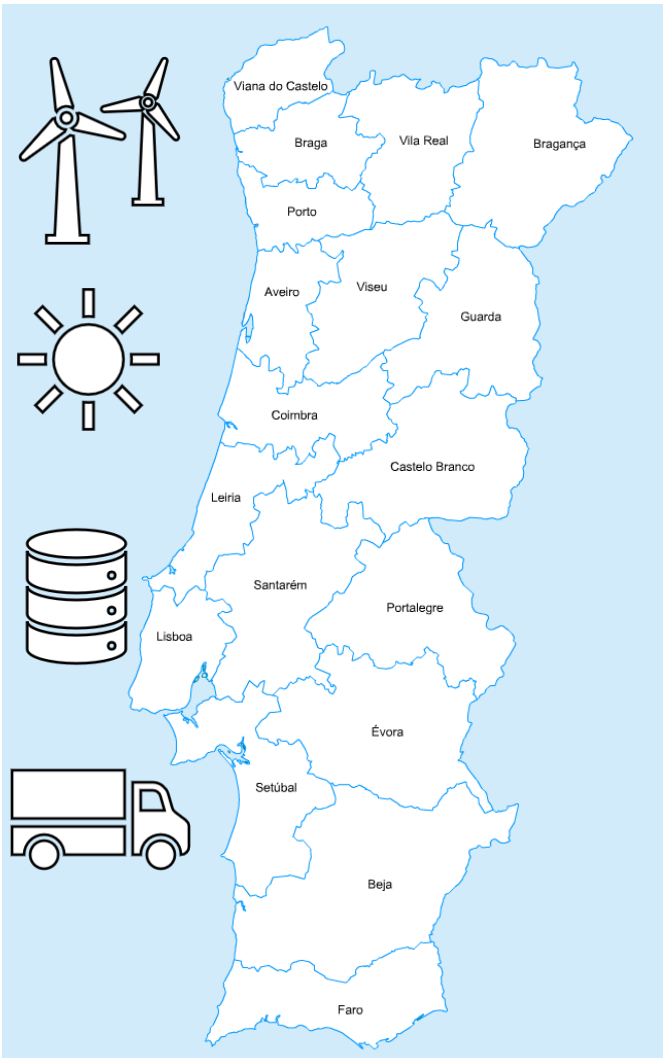
## Decision Variables:

### Continuous variables

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|              |                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------|
| $Pro_{ilt}$  | Amount of H <sub>2</sub> produced at location $i$ by plant of type $l$ at period $t$                                      |
| $Inv_{imt}$  | Inventory level of H <sub>2</sub> at location $i$ in storage type $m$ at period $t$                                       |
| $TC$         | Total SC costs                                                                                                            |
| $TI_t$       | Total capital cost at period $t$                                                                                          |
| $TO_t$       | Total operating cost at period $t$                                                                                        |
| $TT_t$       | Total transportation cost at period $t$                                                                                   |
| $v_{it}$     | Unsatisfied H <sub>2</sub> demand of location $i$ at period $t$                                                           |
| $x_{eijt}^E$ | Flow rate of primary energy source $e$ between grids $i$ and $j$ during period $t$                                        |
| $x_{ijt}^T$  | Amount of H <sub>2</sub> sent from location $i$ to location $j$ via tube trailers at period $t$ for storage purposes      |
| $y_{ijt}^T$  | Amount of H <sub>2</sub> sent from location $i$ to location $j$ via tube trailers at period $t$ for delivery to customers |
| $ESD_{iet}$  | Daily energy source demand of type $e$ at location $i$ at period $t$                                                      |
| $ESC_t$      | Total cost for the energy source consumed for hydrogen production                                                         |

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## Decision Variables:

### Integer and Binary variables

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#### Integer Variables

$n_t^T$

Number of tube trailers/ tanker trucks added to the HSC at period  $t$

#### Binary Variables

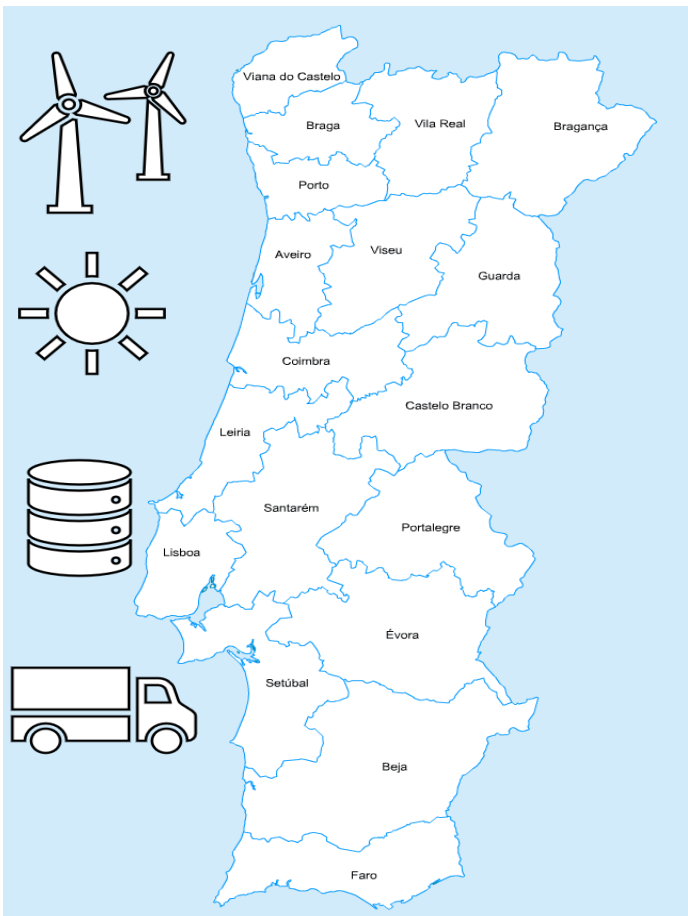
$z_{il}^P$

= 1 if a production plant of type  $l$  is installed at location  $i$

$z_{im}^S$

= 1 if a storage facility of type  $m$  is installed at location  $i$

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## Objective function

The objective function minimizes the total costs, which includes:

- i. In the first part, the fixed costs of opening a production plant/storage facility;
- ii. In the second part, the operating costs for production and storage, transportation and energy costs

$$\min TC = \sum_{t \in T} ( \underset{\substack{\text{Investment Costs} \\ \downarrow}}{TI_t} + \overset{\substack{\text{Operating Costs} \\ \downarrow}}{N(TO_t + \underset{\substack{\text{Transportation Costs} \\ \downarrow}}{TT_t} + \overset{\substack{\text{Energy Costs} \\ \downarrow}}{ESC_t}}) )$$

## Constraints - Mass Balance

### 1st Stage: From Energy Supply to Production Plants

$$1. ESD_{it} = \sum_{l \in L} Pro_{ilt} \gamma, \quad i \in P, t \in T$$

The demand for energy to feed the production is equal to the amount of produced H2 in the production facility  $l$  at location  $i$ , multiplied by the energy conversion factor

$$2. ESD_{it} \leq \sum_{j \in P/\{i\}} \sum_{e \in E} PESAV_{ie} - x_{eijt}^E + \sum_{j \in K/\{i\}} \sum_{e \in E} x_{ejit}^E, \quad \forall i \in P, t \in T$$

The demand for the energy source  $e$  must be covered by local power generation and/or imports from neighbouring grid points.

$$3. \sum_{j \in P/\{i\}} \sum_{e \in E} x_{ejit}^E \leq \sum_{e \in E} PESAV_{ie}, \quad \forall i \in K, t \in T$$

The flow of a certain energy source from location  $i$  to location  $j$  cannot be greater than the energy source availability in location  $i$ .

$$4. \sum_{l \in L} Pro_{ilt} + \sum_{j \in C/\{i\}} (x_{jit}^L + x_{jit}^T - x_{ijt}^L - x_{ijt}^T) = \sum_{m \in M} Inv_{imt}, \quad \forall i \in C, t \in T$$

The amount of H2 that is stored at each period in each location is equal to the amount of H2 produced in that period and location, and/or the inflow and outflow from that location

## Constraints - Mass Balance

### 2nd Stage: From Production Plants to Storage Facilities

$$1. \sum_{l \in L} Pro_{ilt} + \sum_{j \in C / \{i\}} (x_{jit}^L + x_{jit}^T - x_{ijt}^L - x_{ijt}^T) = \sum_{m \in M} Inv_{imt}, \forall i \in C, t \in T$$

The amount of H2 that is stored at each period in each location is equal to the amount of H2 produced in that period and location, and/or the inflow and outflow from that location

$$2. \sum_{j \in S \setminus \{i\}} x_{ijt}^T \leq \sum_{l \in L} Pro_{ilt}, \forall i \in P, t \in T$$

The quantity of shipped H2 by tube trailers/tanker trucks from a production plant cannot exceed the corresponding production amount.



## Constraints - Mass Balance

### 3rd Stage: From Storage facilities to Demand Points

$$1. \sum_{m \in M} Inv_{imt} + \sum_{j \in C/\{i\}} (y_{jit}^L + y_{jit}^T - y_{ijt}^L - y_{ijt}^T) + v_{it} = b_i, \forall i \in C, t \in T$$

The demand at each location in different periods will be met through the amount of stored H2 in that location and/or the inflows and outflows from that location

$$2. \sum_{j \in B/\{i\}} y_{ijt}^T \leq \sum_{m \in M} Inv_{imt}, \forall i \in S, t \in T$$

The amount of H2 shipped by tube trailer/tanker truck from a storage facility cannot exceed the corresponding stored amount of H2

$$3. v_{it} \leq (1 - \alpha_t) b_i, \forall i \in B, t \in T$$

The unsatisfied H2 demand of location  $i$  at period  $t$  must respect the H2 penetration rate for each period

## Constraints – Capacity Constraints

$$1. \text{Pro}_{ilt} \leq \sum_{\substack{t' \in T \\ t' \leq t}} u_l^P, \forall i \in P, l \in L, t \in T$$

The production needs to be lower than the production capacity of the plant

$$2. \text{Inv}_{imt} \leq \sum_{\substack{t' \in T \\ t' \leq t}} u_m^S, \forall i \in S, m \in M, t \in T$$

The same logic is applied to the storage facilities

# GHSC – Galp Project

## Sines Refinery Electrolyser

### Context and Motivation

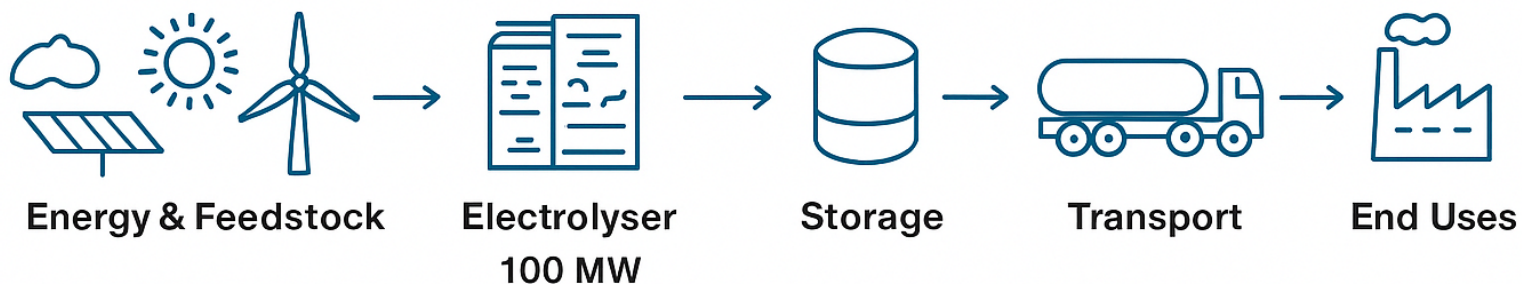
- EU “Fit for 55”, carbon neutrality; >65% GHG reduction;
- Transition from fossil fuels to renewables green hydrogen becomes key;
- IRENA forecasts green/blue H<sub>2</sub> production to rise from 1 Mt/year today to 614 Mt/year by 2050.

### Galp’s Initiative

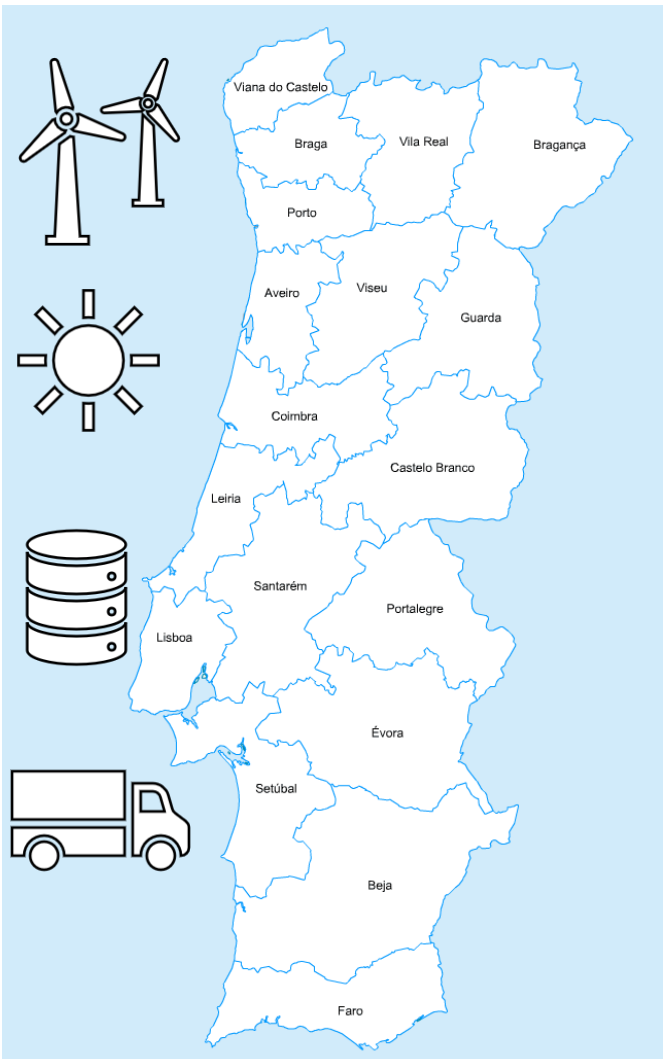
- Strong renewable energy strategy in Iberian Peninsula and Brazil;
- Early steps in green H<sub>2</sub> production, targeting industrial decarbonization in Sines region.
- **Installation of 100 MW PEM electrolyser for green H<sub>2</sub> production**

### Galp’s Initiative

- Develop regional/national GH<sub>2</sub> supply chain models;
- Value chain characterization:
  - Feedstock & energy sources (PPAs, local generation, water treatment);
  - H<sub>2</sub> production: centralized vs decentralized, technology options;
  - Transport: tube trailers / carriers
  - Storage and end uses (industrial feedstock, energy applications);
- **Innovative modeling tool to support planning & decision-making.**



# Case Study: Portuguese Industrial Sector



## Assumptions and Data

- As the focus of this problem is the Portuguese industrial sectors, and to effectively deal with large amounts of data - including suppliers, demand patterns, transportation modes, and costs - data is aggregated by each Portuguese district. Specifically, the logistics network optimization will consider a set of locations corresponding to each of the 18 Portuguese districts

## Renewable Energy Sources and Availability: Wind and Solar Power

- Considering Portugal's abundant natural resources, wind and solar energy are identified as the most efficient sources to feed the electrolysis process;
- In terms of solar resources, Portugal, particularly the southern region, boasts the highest solar radiation in Europe, ranging from 1 500 to 1 900 kWh/m<sup>2</sup>, resulting in higher efficiency for solar energy production compared to other European countries (*Presidência do Conselho de Ministros, 2020*);
- Regarding wind potential, wind energy, harnessed through onshore wind farms, is one of the primary sources of renewable electricity in the national grid, second only to hydropower during years of abundant rainfall (*Direção Geral de Energia e Geologia, DGEG*).

## Renewable Energy Sources and Availability: Wind and Solar Power

- To estimate the availability of renewable energy from wind and solar sources across the national territory, data on the installed capacity of each source by district was first collected. Subsequently, the national production of renewable energy from wind (onshore and offshore) and solar sources was obtained at an annual basis, considering the year 2024. This data, sourced from the Endogenous Energies of Portugal (Energias Endógenas de Portugal “E2P”) data centre;
- Based on the installed capacity at both national and district levels, a weighted percentage (ponderation) for each district was calculated to estimate the distribution of wind and solar energy production across each district. This approach allowed for the estimation of the wind and solar energy availability, in each district, to power the electrolysis production process for green H<sub>2</sub> generation.

## Hydrogen Demand

- In the absence of historical data or relevant background to analyse and estimate the industrial H<sub>2</sub> demand, the consumption of NG from industrial sectors was assumed to be equivalent to the future H<sub>2</sub> demand, as the decarbonization pathway considers the replacement of NG consumption with green H<sub>2</sub>.
- Therefore, the consumption of NG from each of the identified critical industry sectors – refining, chemical, metallurgical, extractive, cement, and glass and ceramics - was collected, at an annual basis (2023), from the Portuguese General Directorate of Energy and Geology (DGEG).
- DGEG's data is available, by industry in each district.



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## Questions and Clarifications

Every Monday and Wednesday, 14:00 – 17:00  
Dep. of Engineering and Management, 2N.8.6  
(Floor 2, Nucleus 8, Office 8.6)

Please note that attendance **is by prior scheduling only.**

You must send me **an email in advance to request a time,** and I will confirm your appointment.

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